

Monthly Intelligence Report

Edition 001 — The Pentalateral CWE Corridor

Where the energy transition is outrunning the grid. Pre-launch · September 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

1,640

528 HV $\geq$ 110 kV · 265 MV (20–110 kV) · 847 distribution-tier

GRID LINES MAPPED

4,190 km

3,984 HV $\geq$ 110 kV · 147 MV · OSM live extraction

MC SIMULATIONS

39.8 M

20,000 per substation · Gaussian copula

PUBLIC DATA SOURCES

28

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 130,000+ source data points per run, executes 39.8 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 0.91 billion arithmetic operations — producing 119,400 output data points with full uncertainty quantification across 1,640 substations (LIVE — OSM) and 4,757 grid lines spanning 4,190 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 28 verified public sources, making the methodology fully auditable and reproducible. Initially covering the Netherlands' entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

380 kV Beverwijk

North Holland, North Holland · 380 kV

0.180

R_FINAL

Low

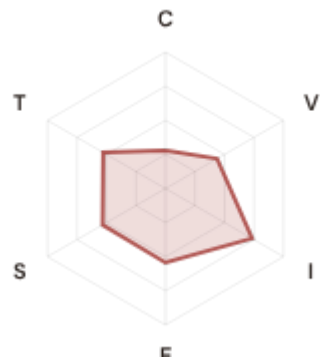
CLASSIFICATION

0.164

CI WIDTH

31th

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.281 / 0.30 (94%)

I Infrastructure: 0.734 / 0.25 (294%) ▲

S Saturation: 0.533 / 0.20 (267%)

V Voltage: 0.437 / 0.10 (437%)

E Economic: 0.544 / 0.10 (544%)

T Transition: 0.528 / 0.05 (1056%)

MODIFIERS

R3 Consequence: 1.111 ↑ amplifies

R4 Graph Criticality: 1.238 ↑ amplifies

R6a Restoration: 0.930 ↓ dampens

R6b Seismic: 1.000 → neutral

R7 Digital Readiness: 1.000 → neutral

This month's spotlight — automatically selected for June 2026 — is **380 kV Beverwijk** in North Holland, North Holland. With an R_final of 0.180 (Low band, 31th fleet percentile), its risk profile is dominated by the T (Transition) component at 1056% of its weight ceiling, followed by E (Economic) at 544%. Modifiers collectively shift R_base by 27.3%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Netherlands Numérique / Netherlands 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

0

High/Critical + R7 < median

AVG R7 MODIFIER

1.0000

Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

410

25.0% · R3 ≥ 1.14

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



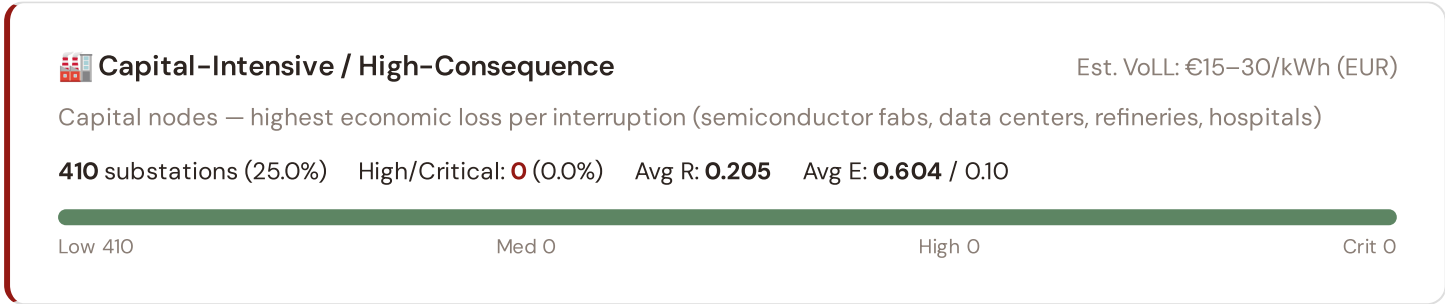
● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **0 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0000$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in —. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Industrial / Medium–Large Enterprise

Est. VoLL: €8–15/kWh (EUR)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

410 substations (25.0%) High/Critical: **2** (0.5%) Avg R: **0.275** Avg E: **0.581** / 0.10



Commercial / SME–Dense

Est. VoLL: €3–8/kWh (EUR)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

406 substations (24.8%) High/Critical: **1** (0.2%) Avg R: **0.345** Avg E: **0.531** / 0.10



Light / Agricultural / Rural

Est. VoLL: €1–3/kWh (EUR)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

414 substations (25.2%) High/Critical: **16** (3.9%) Avg R: **0.394** Avg E: **0.464** / 0.10

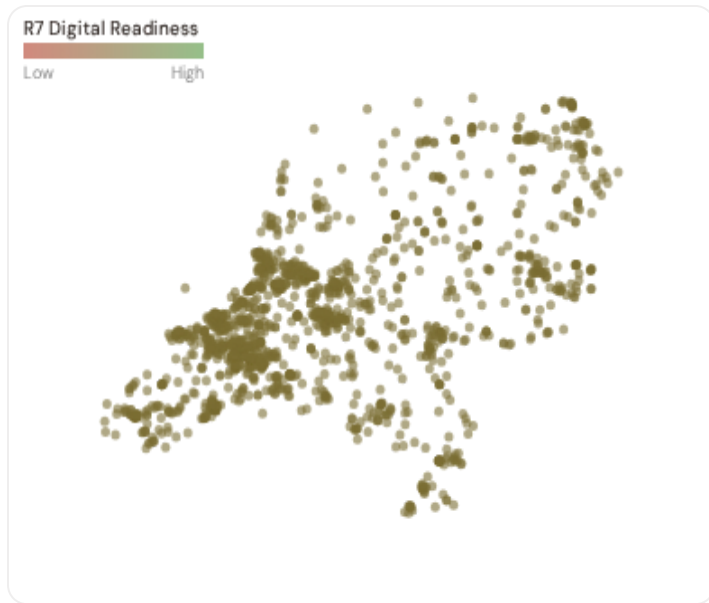


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **0 substations** combine capital-intensive economic fabric ($R3 \geq 1.14$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **South Holland (291), North Holland (107), Utrecht (12)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 4.9%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Province ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify Province combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Drenthe ▲ high R		1.0000
2	Flevoland		1.0000
3	Friesland		1.0000
4	Gelderland		1.0000
5	Groningen ▲ high R		1.0000
6	Limburg (NL) ▲ high R		1.0000
7	North Brabant ▲ high R		1.0000
8	North Holland ▲ high R		1.0000
9	Overijssel ▲ high R		1.0000
10	Utrecht		1.0000
11	Zeeland		1.0000
12	South Holland		1.0000

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Drenthe** has the lowest average R7 (1.0000), while **South Holland** leads at 1.0000 — a spread of 0.0000 across the R7 range. **0 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7’s dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0000, median = 1.0000; 0 substations classified HIGH cyber-exposure; 0 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **28 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI’s credibility.

TOTAL SOURCES

28

95 variables

WEEKLY / MONTHLY

1

High-frequency feeds

QUARTERLY / ANNUAL

18

Regular refresh cycle

STATIC / DERIVED

6

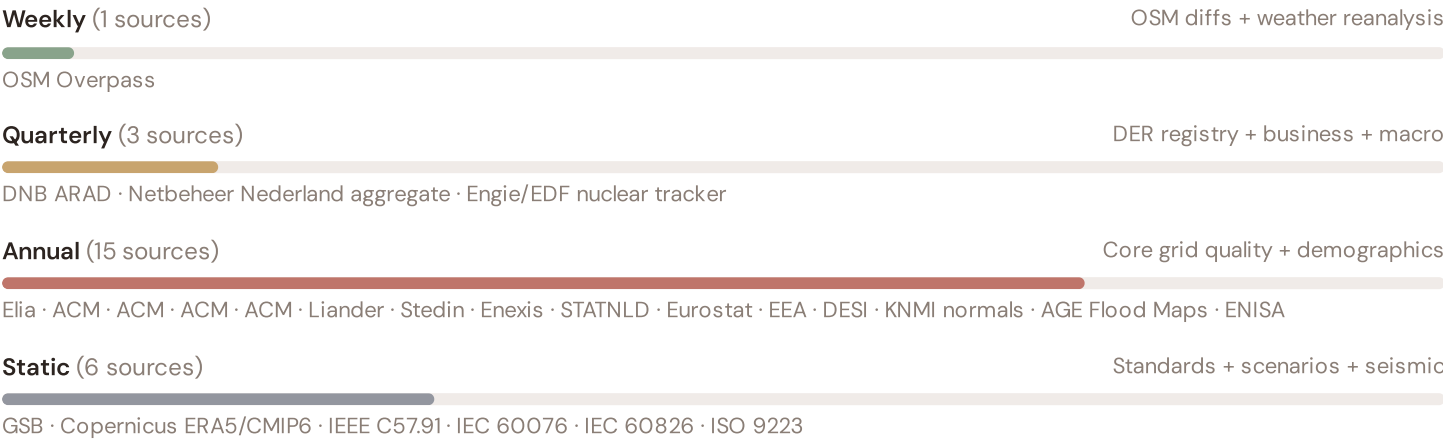
Data Source Registry — June 2026

OSM	OpenStreetMap Power Infrastructure	CONTINUOUS	Node / edge	4 vars
SYNERGRID	Netbeheer Nederland — Federated DSO aggregate (11 DSOs)	QUARTERLY	National	3 vars
NUCLEAR	Engie + EDF Netherlands — nuclear phase-out tracker	QUARTERLY	NPP	2 vars
DNB	DNB — Banque nationale de Belgique	QUARTERLY	NUTS2	2 vars
ACM	ACM — Federal Quality of Supply Reports	ANNUAL	National	5 vars
ACM	ACM — Flemish regional regulator	ANNUAL	Region	3 vars
CWAPE	ACM — Walloon regional regulator	ANNUAL	Region	3 vars
ACM	ACM — Brussels regional regulator	ANNUAL	Region	2 vars
FLUVIUS	Liander — Flemish single-DSO annual report	ANNUAL	DSO area	4 vars
Stedin	Stedin — Walloon dominant DSO annual report	ANNUAL	DSO area	4 vars
SINLDGA	Enexis — Randstad core DSO annual report	ANNUAL	DSO area	4 vars
STATNLD	STATNLD — Statistics Netherlands	ANNUAL	Commune	8 vars
EUR	Eurostat — Energy Statistics (BE NUTS2 cuts)	ANNUAL	NUTS2	3 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km	2 vars
DESI	DESI Digital Economy & Society Index	ANNUAL	EU Regional	1 var
GSB	GSB — Geological Survey of Netherlands + EMSC	STATIC	~5 km	2 vars
AGE-FL00D	Administration générale Eau — Flood maps	STATIC	~250 m	1 var
CATTENOM	Cattenom NPP (EDF FR) — cross-border proximity	STATIC	Province	1 var
IEEE-1	IEEE C57.91 Thermal Model	STATIC	Asset	1 var
IEC-1	IEC 60076 Power Transformers	STATIC	Asset	1 var
IEC-826	IEC 60826 Overhead Line Loading	STATIC	Zone	1 var
ISO-9223	ISO 9223 Atmospheric Corrosion	DERIVED	Province	1 var

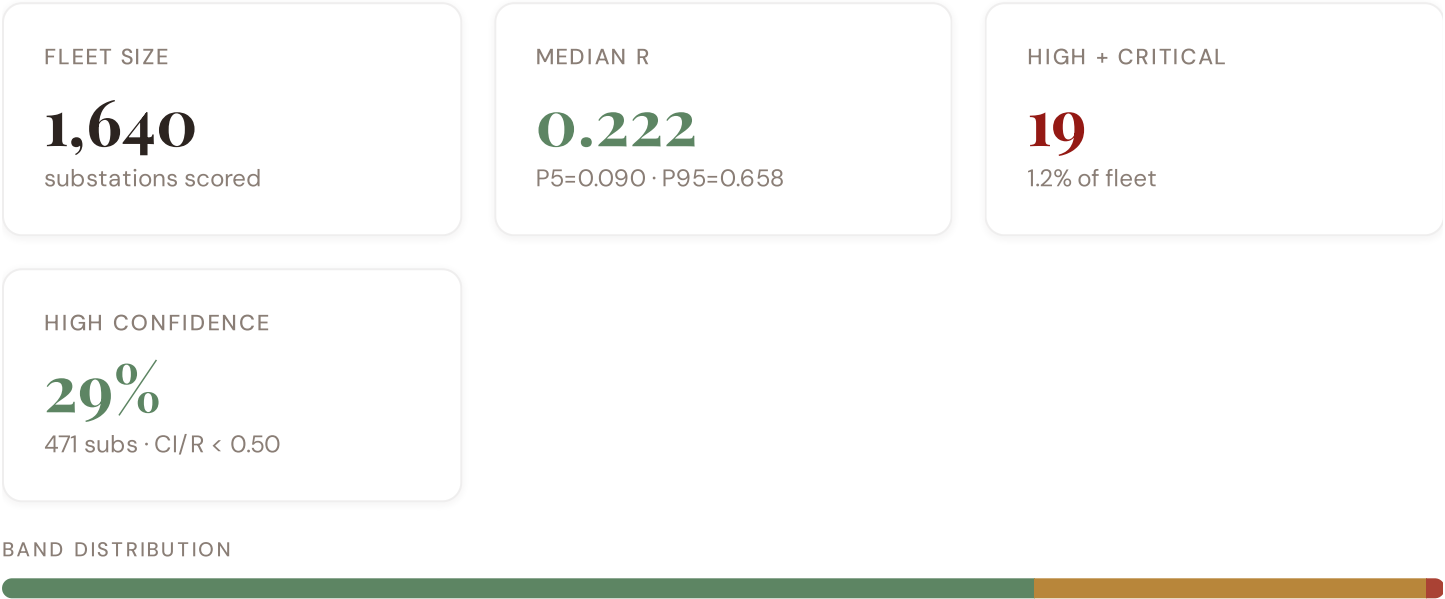
ELIA	TenneT — TSO annual + 10-min telemetry	ANNUAL + 10-MIN	National + zone	8 vars
EPEX	EPEX SPOT — BE day-ahead clearing	HOURLY	Zone	3 vars
ENTSOE	ENTSO-E Transparency — NL bidding zone	HOURLY	Bidding zone	4 vars
RMI	KNMI / IRM — Royal Meteorological Institute	HOURLY + 30Y NORMS	Station	5 vars
CDS	Copernicus CDS / ERA5 + CMIP6 SSP2-4.5	HOURLY + STATIC	0.25° (~25 km)	4 vars
OPM	Open-Meteo Weather API	HOURLY	~1 km	1 var
ENISA	ENISA — EU cybersecurity index	BIENNIAL	EU country	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS



C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **1,640 substation scores remain unchanged** this edition. The fleet median R of 0.222 (mean 0.305) reflects a distribution skewed toward the lower bands, with 71.6% in Low and 27.3% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

19 changes tracked across PO, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	\$2, \$4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	\$5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio = STATNLD EU-SILC)	\$5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection (NetworkX)	\$5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	\$5
F6	ENHANCED	R7 Digital Readiness — Province-level DESI model	\$5
BE1	NEW	Netherlands onboarded — 11 source modules (incl. d02b_netbeheer for multi-DSO); d05_osm LIVE 1,640 substations P2	Stage 1
BE2	NEW	BE regulatory stack mapped: TenneT / Netbeheer Nederland (Liander+Stedin+Enexis+8 IMs) / ACM + ACM/ACM/ACM P2	Stage 1
BE3	NEW	Scoring engine: 12 provinces × 6 components → R_final + Markov ETTC + 20k MC + Sobol P2	Stage 2
BE4	NEW	Dashboard: 8 HTML pages + Pattern C ssi-metadata.js + 28-source registry P2	Stage 3
BE5	ENHANCED	R3 economic tier thresholds BE-calibrated [1.066 / 1.053 / 1.027] from actual distribution P2	Stage 4
BE6	ENHANCED	R6b seismic α band BE [0.10, 0.30] — wider than LU [0.05, 0.20] (Groningen induced) P2	Stage 4
BE7	ENHANCED	Multi-DSO SAIDI weighting per province (Liander/Stedin/Enexis) via d02b_netbeheer P2	Stage 4
BE8	ENHANCED	Markov rates BE-older calibrated (λ_12=0.045, λ_23=0.07, λ_34=0.11) P2	Stage 4
BE9	NEW	Automation pipeline scaffolded — FIRST_REFRESH=2026-09 P2	Stage 5
BE10	NEW	Landing-page Netherlands map-active (oecd → active flip) + OECD count update 25 → 26 P2	Stage 6
D1	DATA	28-source BE registry verified (vs LU 22, CZ 30, IT 30) — all open-licensed	Date

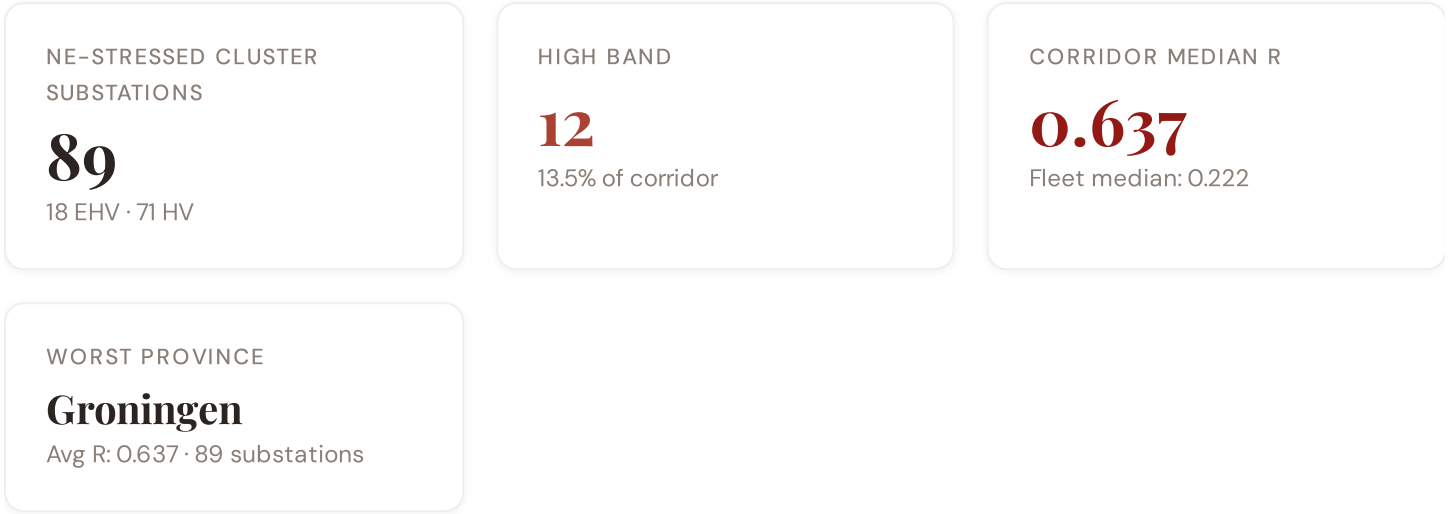
D2	DATA	OSM Overpass live extraction: 1,640 substations + 4,757 power lines + 4,190 km (area filter)	Date
D3	DATA	KB §44 — d05_osm patched: bbox → area[ISO3166-1=BE] + point-in-polygon (geoBoundaries ADM2). Dropped 770 cross-border subs. P2	Date

SECTION D — MONTHLY DEEP-DIVE

The Pentalateral CWE Corridor: Where Coal Reliance Meets Energy Transition Pressure

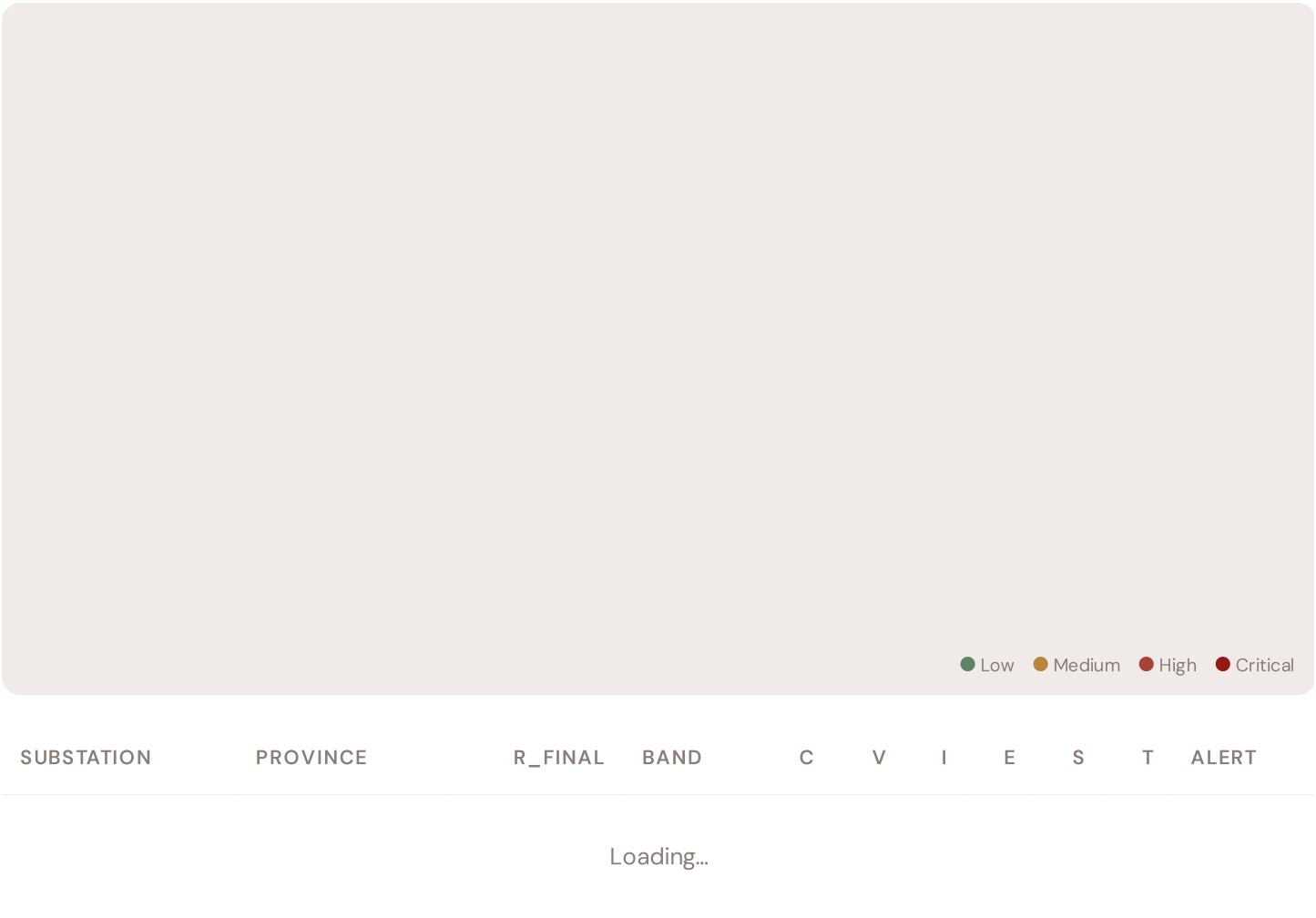
Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** NE-stressed cluster · **002** DER saturation hotspots (border provinces) · **003** Climate vulnerability under CMIP6 · **004** Urban-Rural economic impact disparity · **005** coal-fleet decommissioning risk and transition risk · **006** Infrastructure age vs smart-meter rollout · **007** Stress-Resilience quadrant trends · **008** T-component forward projections · **009** Province Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the NE-stressed cluster, Why Now



The Groningen hosts **89 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **12 substations (13.5%)** are classified High, with only **0** in the Low band. 100% of corridor substations score above the national fleet median of 0.222. The corridor median R of 0.637 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



D.3 Component Analysis

The dominant risk driver across the Groningen corridor is **T (Transition)**, averaging 0.499 against a fleet average of 0.501 — occupying 998% of its weight ceiling ($w = 0.05$). The second contributor is **V (Voltage)** at 0.607 vs fleet 0.458 (607% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.057 (fleet: 1.112), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 0.999, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

The region regions-level breakdown reveals significant intra-regional variation. **Groningen** leads with a mean R of 0.637 across 89 substations, while **Groningen** scores 0.637 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Groningen is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

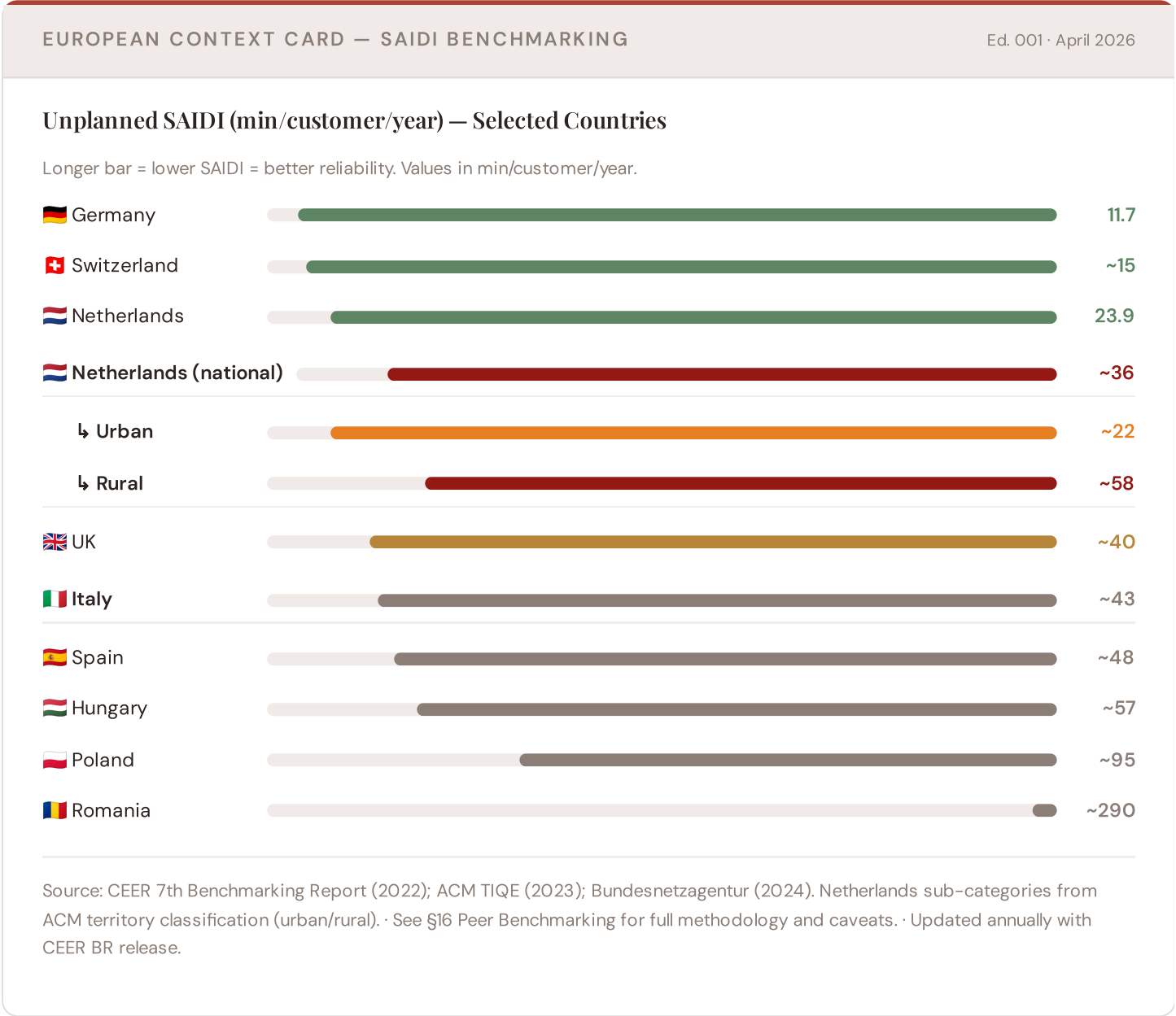
36 Groningen substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Groningen sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are

disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the NE-stressed cluster analysis hinges on the Netherlands' continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.



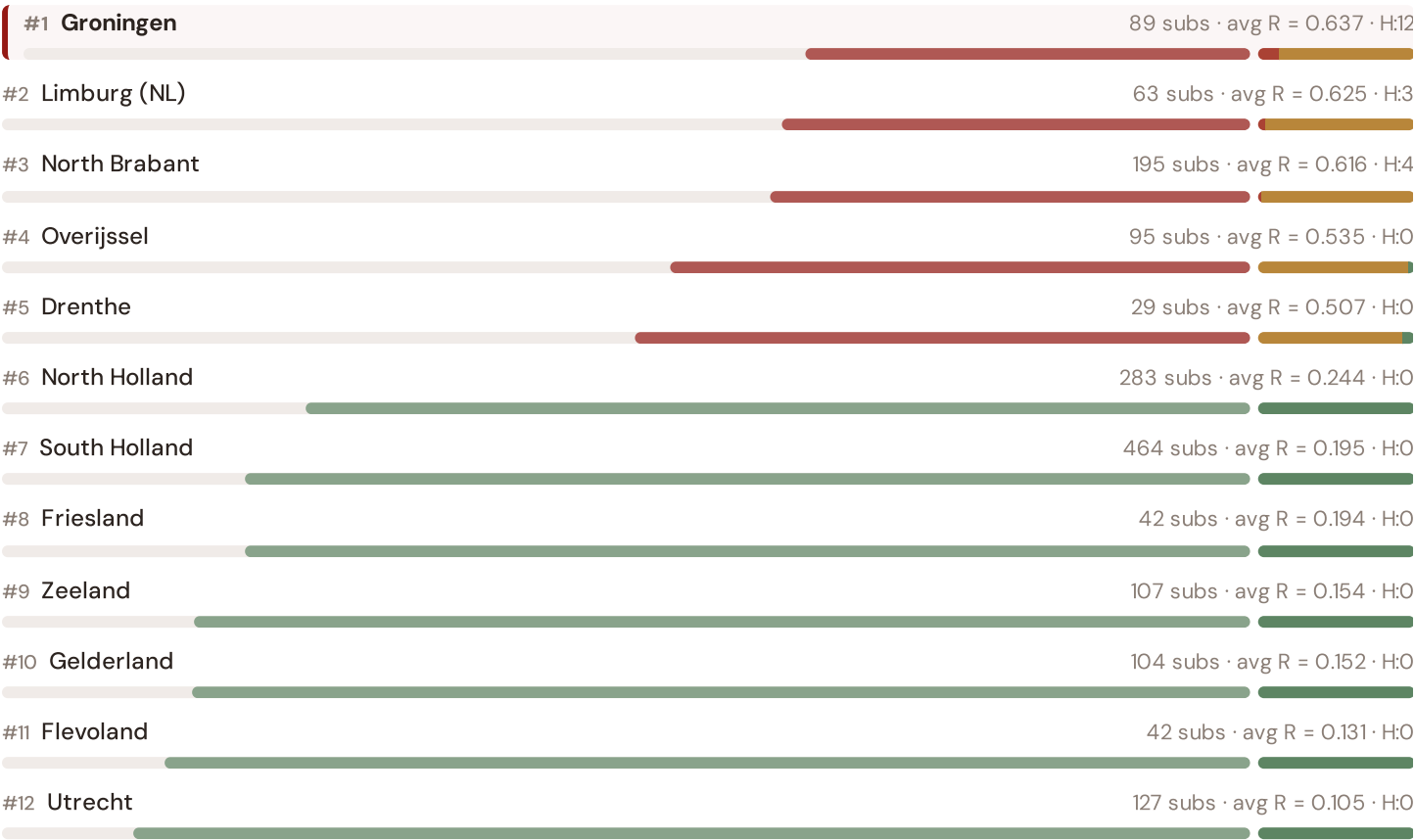
This month's key insight: The Groningen corridor deep-dive shows **89 substations** (of 89) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The

corridor's average C of 0.897 is **2.3× the fleet average** of 0.382. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 13 LU Provinces

Where does the NE-stressed cluster sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

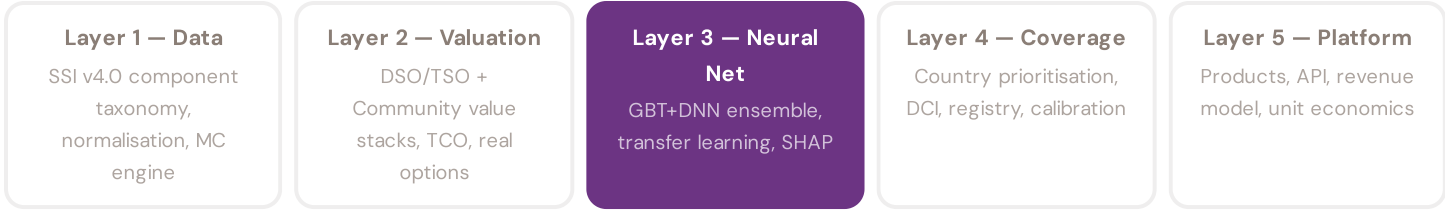


The Groningen ranks **#1 of 12 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Groningen, Limburg (NL), North Brabant, while the three most resilient are Utrecht, Flevoland, Gelderland. 5 of 12 score above the fleet average of 0.305. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the TenneT + Netbeheer Nederland/TenneT + Netbeheer Nederland and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 1,640 substations with complete feature vectors. Average CI_width of 0.164 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 1174 Low, 447 Medium, 19 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 03

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 3 Quarterly Sources

DNB ARAD, Netbeheer Nederland aggregate, Engie/EDF nuclear tracker are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **STATNLD — Statistics Netherlands** (8 variables → P1 pop density, EP_rate ~7.8% national, elderly, healthcare; per-province + per-commune).

FLEET SPOTLIGHT

Groningen: Highest Risk Concentration

Groningen has the highest concentration of High/Critical substations: **12 of 89 (13%)** are in the upper risk bands — avg R = 0.637. This makes it the most acute risk corridor in the fleet.

Edition 03 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Netherlands utility, ACM, TenneT + Netbeheer Nederland, TenneT + Netbeheer Nederland, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Pre-launch placeholder · First refresh 2026-09-10 · SSI Index v4.0.2 · 1,640 substations (LIVE OSM) · 4,757 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5